GUIDE TO USE IV INFUSION HEPARIN PROTOCOL FOR VENOUS THROMBOEMBOLISM (VTE) TREATMENT

STEP 1 : IDENTIFY HEPARIN PROTOCOL TO USE

Identify patient's weight, height, age & calculated BMI.

- Weight $\leq$ 60kg
- BMI $\geq$ 30 kg/m^2
- Age $\geq$ 70 years old

Yes

Use Option A
(weight-based)
FOR UNDERWEIGHT, OVERWEIGHT or ELDERLY PATIENTS ONLY

No

Use Option B
(non weight-based)
ALL OTHER PATIENTS

STEP 2 : CALCULATE HEPARIN INFUSION DOSE

Identify correct *Unfractionated Heparin Continuous Infusion Protocol* to use.

Calculate patient's IVI Heparin dose according to steps within each protocol.

UNFRACTIONATED HEPARIN CONTINUOUS INFUSION PROTOCOL

OPTION A : FOR UNDERWEIGHT, OVERWEIGHT or ELDERLY PATIENTS ONLY (weight-based)

STEP 1: INITIATION OF HEPARIN

Table 1: Treatment dose of heparin initiation

*BMI ≥40 kg/m^2 = use *adjusted* body weight

a) Age <70 yo = Initial slow bolus **80 unit/kg (undiluted)** over at least 1 min (max dose 10,000 units)

Age ≥70yo = Initial slow bolus **50 unit/kg (undiluted)** over at least 1 min (max dose 10,000 units)

Followed by:

b) IV infusion **18 units/kg/hour** maintenance dose
(maximum 2,000 units/hour)
(dilution as below)

STEP 2: PREPARATION AND ADMINISTRATION

a) Initial slow bolus
- Undiluted

b) IV continuous infusion
- Dilute 25,000 units (5ml of Inj Heparin **5000iu/ml**) up to 50ml of NS/D5% in 50cc syringe
- 1ml = 500 units

*Patient should have a dedicated line for heparin infusion.

STEP 3: MONITORING

APTT blood samples **CANNOT** be drawn from the same line.

For baseline APTT prior to IV Heparin infusion initiation.
Monitor APTT every 6 hours until target APTT (46 – 70 sec) is achieved.
Then, monitor APTT according to Table 2.
(Result to be reviewed by MO / Specialist immediately once received)

STEP 4: DOSING ADJUSTMENT

Review APTT and adjust infusion rate if required.

Table 2 : IVI Heparin Dose Adjustment

APTT	Action and Dose Change	Next APTT
<35s	Give bolus = 80 unit/kg STAT *(age≥70yo = 50unit/kg),* New maintenance infusion rate = Current infusion rate **+ 4 unit/kg/hr** Continue with this new infusion rate until next APTT sampling.	6 hour
35 to 45s	Give bolus = 40 unit/kg STAT *(age≥70yo = 25unit/kg),* New maintenance infusion rate = Current infusion rate **+ 2 unit/kg/hr** Continue with this new infusion rate until next APTT sampling.	6 hour
46 to 70s	No change. Maintain current dose of IV continuous infusion rate.	24 hour
71 to 90s	Adjust to new maintenance infusion rate = Current infusion rate **- 2 unit/kg/hr** Continue with this new infusion rate until next APTT sampling.	6 hour
>90s	**Withold infusion for 1 hour**, New maintenance infusion rate = Current infusion rate **- 3 unit/kg/hr** Continue with this new infusion rate until next APTT sampling.	6 hour

OPTION B : ALL OTHER PATIENTS (non weight-based)

STEP 1: INITIATION OF HEPARIN

Table 3: Treatment dose of heparin initiation

a) Initial slow bolus **5000 IU (undiluted)** over at least 1 min

Followed by:

b) IV continuous infusion **1000 units/hour** maintenance dose (dilution as below)

STEP 2: PREPARATION AND ADMINISTRATION

c) Initial slow bolus
- Undiluted

d) IV continuous infusion
- Dilute 25,000 units (5ml of Inj Heparin **5000iu/ml**) up to 50ml of NS/D5% in 50cc syringe
- 1ml = 500 units

*Patient should have a dedicated line for heparin infusion.

STEP 3: MONITORING

APTT blood samples **CANNOT** be drawn from the same line.

For baseline APTT prior to IV Heparin infusion initiation.
Monitor APTT every 6 hours until target APTT (60 – 85 sec) is achieved.
Then, monitor APTT according to Table 4.
(Result to be reviewed by MO/ Specialist immediately once received)

STEP 4: DOSING ADJUSTMENT

Table 4 : IVI Heparin Dose Adjustment

APTT	Action and Dose Change	Next APTT
<50s	Give bolus = **5000 unit** STAT, New maintenance infusion rate = Current infusion rate **+ 100 unit/hr** Continue with this new infusion rate until next APTT sampling.	6 hour
50 to 59s	Adjust to new maintenance infusion rate = Current infusion rate **+ 100 unit/hr** Continue with this new infusion rate until next APTT sampling.	6 hour
60 to 85s	No change. Maintain current dose of IV continuous infusion rate.	Next morning blood taking
86 to 95s	Adjust to new maintenance infusion rate = Current infusion rate **- 100 unit/hr** Continue with this new infusion rate until next APTT sampling.	6 hour
96 to 120s	**Withold infusion for 30 minutes**, Adjust to new maintenance infusion rate = Current infusion rate **- 100 unit/hr** Continue with this new infusion rate until next APTT sampling.	6 hour
>120s	**Withold infusion for 60 minutes,** Adjust to new maintenance infusion rate = Current infusion rate **- 200 unit/hr** Continue with this new infusion rate until next APTT sampling.	6 hour

HEPARIN ANTIDOTE: PROTAMINE SULFATE

- Use Protamine for heparin neutralization.
- Heparin concentration decreases rapidly upon discontinuation of heparin infusion (half life of heparin ~60 – 90 minutes). However, if an immediate effect is required, consider administration of protamine sulfate.

STEP 1 : DOSE OF IV PROTAMINE SULFATE

Identify total heparin dose administered in the preceding 2 to 3 hours.
Then calculate dose of protamine sulphate according to the table below;

Time since last dose of heparin	Dose of Protamine (max 50mg/dose)	
Immediate	1mg / 100 units Heparin received	*(or 25mg fixed dose)*
30 minutes – 2 hours	0.5mg / 100 units Heparin received	*(or 10mg fixed dose)*
> 2hours	0.25mg / 100 units Heparin received	*(or 10mg fixed dose)*

STEP 2 : PREPARATION AND ADMINISTRATION

- Administered by slow IV bolus over 10 minutes, at a rate not exceed 5mg/min.
- Rapid administration of protamine sulfate can cause severe hypotensive and and anaphylactoid-like reactions.
- Reconstitution : Not required
- Further Dilution : Not required

STEP 3 : MONITORING

- Monitor APTT : 5 – 15mins after initial dose and then at 2 – 8 hours.

IV INFUSION HEPARIN SODIUM CHART

1 Indication: ______________________________

Target APTT: ______________________________

APTT (just prior to Heparin initiation): __________ secs

Date/Time taken: __________________

PATIENT INFORMATION

Name: ____________

Ward: ____________

SB: ____________

2 Date of Heparin initiation:

____

Body weight (kg) :

____

Initial bolus dose :

________________ units

*Initial infusion dose: (repeat APTT in 6 hours)

________________ units/hour

Time bolus dose given:

Dr's name

Signature

SN's name

Signature

3. Heparin Infusion Prescription			4. Heparin Monitoring				5. Heparin Dose Adjustment (refer Table 2)				
Date / Time of Infusion	Current Maintenance Dose		APTT Blood Result				New Adjusted Dose			Dr's Signature	SN's Signature
	Dose (units/hr)	Rate (ml/hr)	Date	Time coagulation profile taken	APTT result (secs)	Time result reviewed	Bolus Dose (units)	IVI Dose (units/hr)	IVI Rate (ml/hr)		
Examples: 06/05/21 (08:00 AM)	*900*	*1.8*	*06/05/2021*	*2:00 PM*	*40*	*2:00 PM*	*2000*	*1000*	*2*	*Aaaaa*	*Bbbb*

Appendix 2 : Formulae for calculation

Ideal Body Weight (IBW)

IBW (male) =	{[(Ht-150)/30] x 12 x 2.3} + 50
IBW (female) =	{[(Ht-150)/30] x 12 x 2.3} + 45.5

Weight (kg) ; Height(ht) (cm)

Adjusted Body Weight (ABW)

ABW =	IBW + [0.25 x (Current BW – IBW)]

Weight (kg) ; Height (cm)

Body Mass Index (BMI)

BMI =	Weight(kg) / $[\text{Height(m)}]^2$

Weight (kg) ; Height (m)

Reference:

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- Clinical Practice Guidelines: Prevention and Treatment of Venous Thromboembolism. Malaysia: Ministry of Health Malaysia; 2013.
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- Heparin [Internet]. UpToDate. 2021. Available from: http://www.uptodate.com
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- Heparin. Dilution Guide for High Alert Medications: Pharmaceutical Services Division; 2011.